

Forecasting with a Fixed Evidence Set: A Three-Model Comparison against Prediction-Market Probabilities

Vorhersage experimental series

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Abstract

We compare three frontier models on six prediction questions, holding the question definitions, evidence packets, elicitation prompts, and market benchmarks fixed. Thirty-six new calls cross three models with question-only and outside-research conditions. All eighteen research forecasts pass validation. Their mean absolute distances from the original Kalshi midpoints are 5.42 percentage points for Claude Fable 5.1, 6.25 for GPT-6 Astra, and 8.42 for Gemini 3.1 Pro. The original Gemini run yielded 8.58 points. Research reduces average distance for every model on the three contracts with accepted responses in all six arms. These observations illustrate a controlled comparison of forecasting methods using timely market benchmarks; the small sample and single draw per cell do not establish a general ranking of forecasting ability.

1 Design

The original study selected six contracts spanning weather, employment, monetary policy, space technology, entertainment, and baseball. Its outside evidence came from primary sources rather than Kalshi or other prediction markets. This follow-up retains those exact questions and evidence packets. Three models each receive the six questions twice: once without current research and once with the corresponding frozen packet. The calls are independent; a research-condition call does not see a question-only forecast. All 36 returned system and user prompt receipts exactly match their respective prompts from the original study.

Prediction-market probabilities provide informative intermediate evaluation targets and immediate feedback while event outcomes remain unavailable. Evidence for their forecasting value and calibration, subject to liquidity and horizon, is discussed in the companion report. We measure agreement with the same opening midpoints used there, without downloading new prices or outcomes.

The information cutoff remains September 18, 2026, 11:19 UTC. New inference occurred later that evening Eastern time (September 19 UTC). The coordinator had already seen the original prices and forecasts; new model contexts contained neither. These are therefore registered, frozen-data follow-up runs rather than a new fully blind prospective study. The same-day weather outcome may have become known by the later execution time. The original advantage of evaluating genuinely future events before their outcomes exist should not be attributed unconditionally to that later weather call.

2 Evaluation

Let p_{imc} denote the forecast for question i , model m , and information condition c , and let q_i denote the original market midpoint. On a common eligible cohort S , the principal statistic is

$$\text{MAE}_{mc} = \frac{100}{|S|} \sum_{i \in S} |p_{imc} - q_i|.$$

We also report root mean squared distance and preserve distances outside each original bid–ask spread. Research improvement for an individual question is $100(|p_{im0} - q_i| - |p_{im1} - q_i|)$ percentage points. Its relative improvement divides that change by the original absolute distance. A negative value denotes deterioration. These are market-agreement statistics, distinct from outcome scores. Under the stronger assumption that q_i is the conditional event probability for the reference information set, squared distance also equals conditional expected excess Brier loss; marginal calibration alone does not establish that assumption.

3 Results

Table 1 uses all six questions because every research forecast passed validation. Claude is closest on this sample, followed by Astra and Gemini. The new Gemini mean closely resembles its earlier mean, although question-level estimates change. With only one new draw per cell, this is a replication observation rather than a precise estimate of stochastic variability. Excluding same-day weather leaves research MAE of 5.60 points for Claude, 6.20 for Astra, and 9.80 for Gemini on the same five remaining contracts.

Table 1: Research forecasts on the same six contracts. Distances are percentage points. Cost includes both information conditions and rejected responses.

Model	MAE	RMSE	Cost (USD)
Claude Fable 5.1	5.42	6.99	0.84760
GPT-6 Astra	6.25	8.68	0.39430
Gemini 3.1 Pro, new	8.42	11.70	0.17177
Gemini 3.1 Pro, original	8.58	11.40	0.16835

The unchanged conservative text rule excluded three question-only responses: Claude on employment and weather, and Gemini on Netflix. Each mentions prediction markets and infers information from contract selection or threshold placement, despite instructions treating contract rules as definitions only. Claude’s employment response also includes an explicit disclaimer of using odds; a mere mention is not proof of price exposure. These responses remain excluded without repair. Acceptance is 12/12 for Astra, 10/12 for Claude, and 11/12 for Gemini.

The fully matched before/after cohort consequently contains Fed, Neutron, and baseball. Table 2 compares identical questions in all arms. Dropping same-day weather, as prespecified, does not change this cohort. Available-case results are archived separately rather than substituted for matched comparisons.

Figure 1 preserves the question-level comparison. Effects are heterogeneous: Claude’s Netflix forecast moves slightly away from the midpoint, and Gemini’s Fed forecast does not move. The unchanged Netflix packet also retains incomplete release context from the original study; replacing it would constitute a different data experiment. A larger repeated comparison could separate model variation from evidence-quality effects.

Table 2: Fully matched comparison on three contracts.

Model	Question-only MAE	Research MAE	Reduction
GPT-6 Astra	18.00	7.67	57.4%
Claude Fable 5.1	17.00	4.00	76.5%
Gemini 3.1 Pro, new	30.33	8.00	73.6%

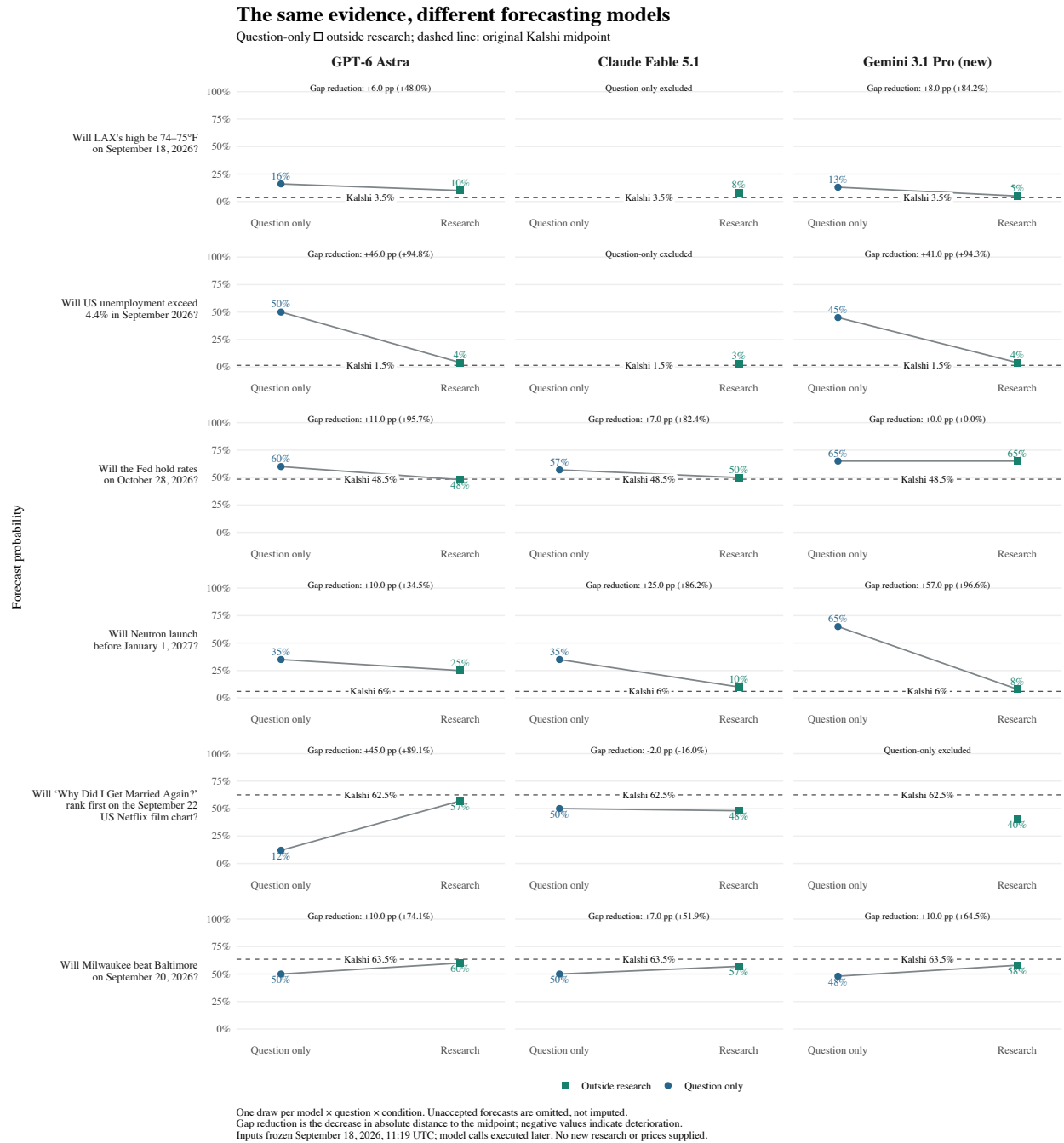


Figure 1: Fixed-data model comparison. Slopes connect independent question-only and research forecasts. Dashed lines mark the original Kalshi midpoints. Annotations report absolute and relative gap reductions. Excluded responses have no point or connecting segment; their accepted counterpart is retained.

A Implementation and reproducibility

Vorhersage separates deterministic record keeping from external model judgment. The experiment registers six immutable arms, each combining the same direct-judgment method with a specific model and evidence condition. Frozen packets carry source provenance and content-derived identifiers. Every trial has independent assessment and issuance state; deterministic issuance requires no additional model call. Runs are marked retrospective and preserve actual execution times rather than backdating forecasts to the information cutoff.

Expected Parrot transports one fresh, tool-free EDSL interview per cell. Astra uses high reasoning effort, Claude adaptive thinking at high effort, and Gemini the original 2,048-token thinking budget and temperature 0.5. All have an 8,192-token output ceiling. These are model configurations rather than matched computation budgets. Provider adapters may omit serialized legacy defaults; the archive verifies returned model identities and prompt receipts without claiming independent attestation of server-side settings or metering.

Import validates scenario, model, agent, JSON schema, evidence references, and the excluded-content rule. An initial importer defect compared plain scenario dictionaries with EDSL's serialized dictionaries and rejected all rows. The corrected validator requires the two exact serialization metadata fields and unchanged substantive content. Initial rejection records are retained. Neither forecasts nor registered inputs changed, and no additional inference occurred.

Reported new response costs total \$1.41367, including excluded responses; all costs are known in the returned records. Provider job summaries report twelve completed interviews and no interviews with exceptions for each model. Source collection and coordination costs are outside this total. Raw responses, model settings, prompt hashes, submissions, acceptance records, issued forecasts, and seal commitments are archived under `run/`.

The comparison script is `compare.py`. A separate `ggplot2` script, `plot_performance.R`, generates the figure. Running `verify`, `audit`, and `score` commands rechecks inputs and reconstructs the comparison without new inference or web access. The new boundary tests cover unchanged prompts, serialization metadata, unexpected context, changed model settings, and retention of the strict output exclusion policy. No dependency is added to the core package.